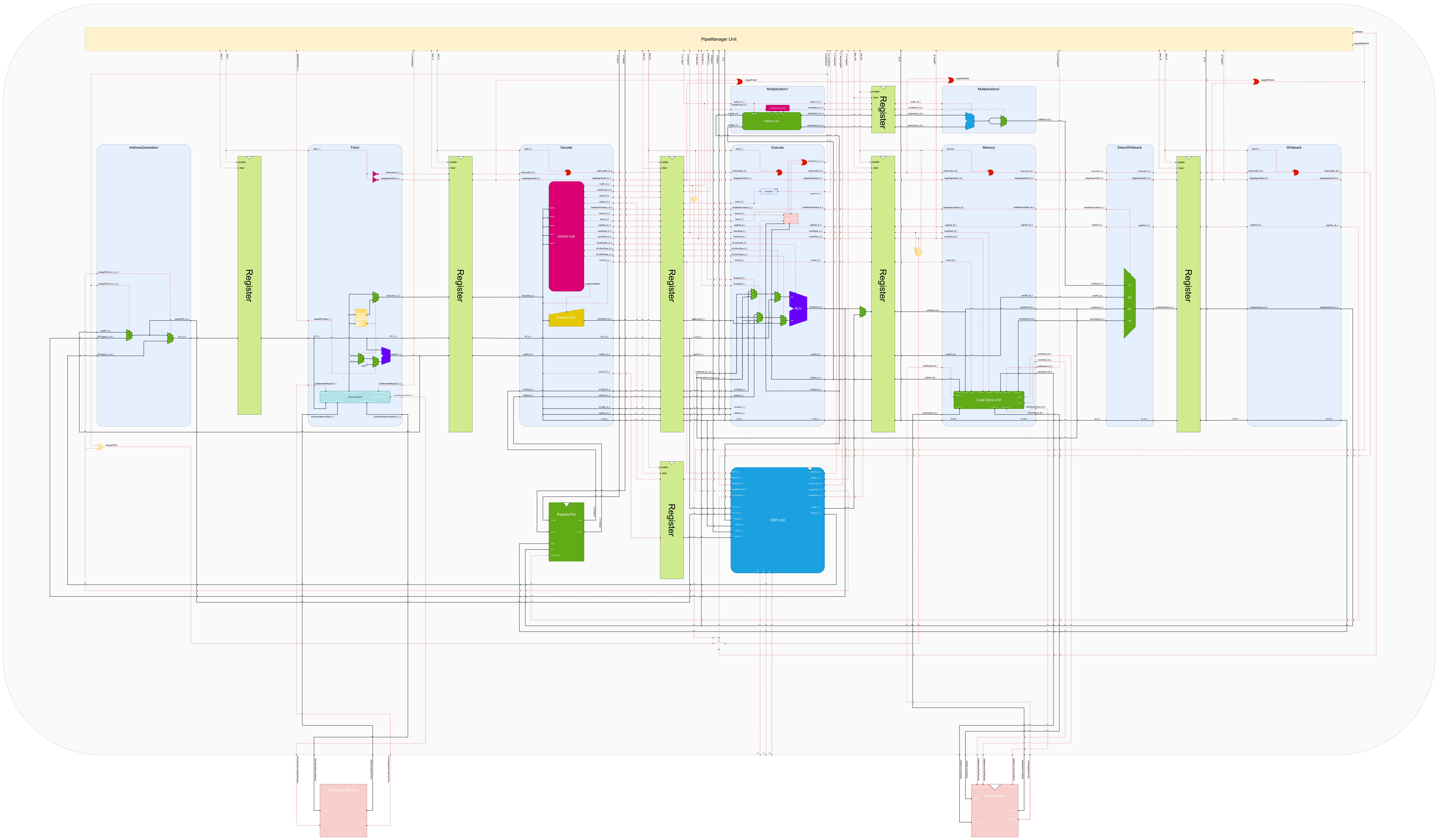
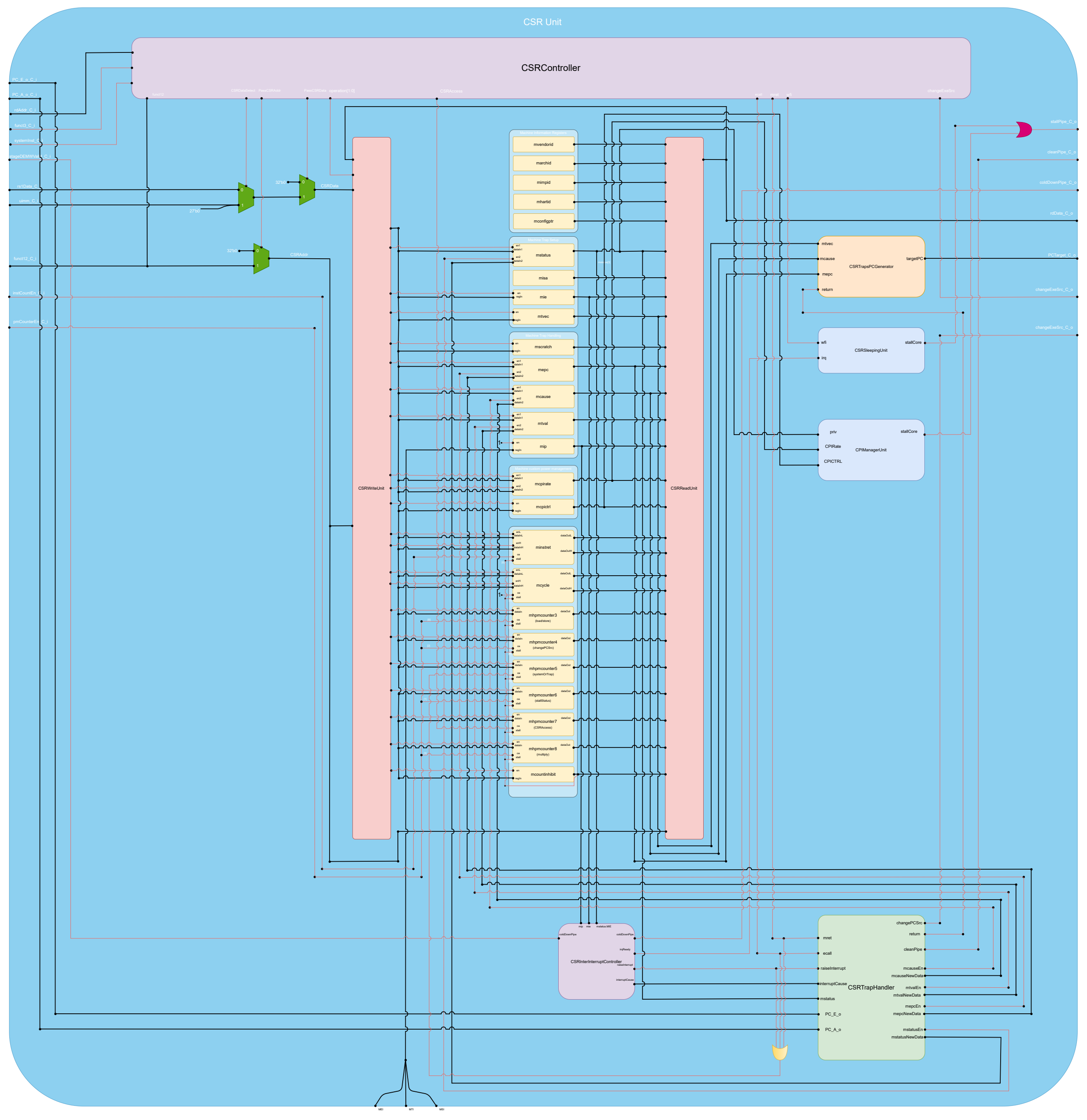
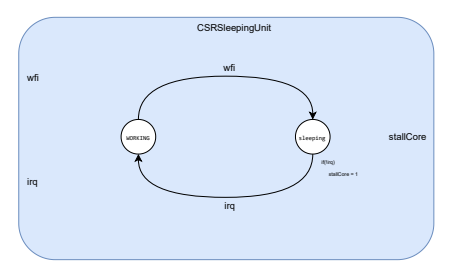
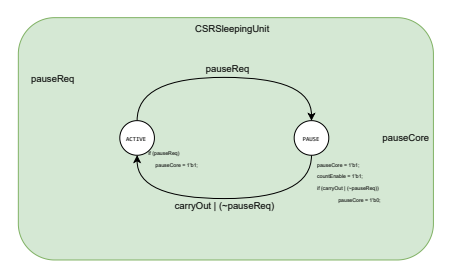

PipeManager Unit
stallStatus
stageDEMWValid
clear_F
stall_F
disablePCAdder_F_i
waitRequest_F_o
clear_D
stall_D
rs1Add_D
rs2Add_D
clear_EC
stall_EC
mulEn_X1_i
memRead_E
memWrite_E
forwardB_E
forwardA_E
rs1Add_E
rs2Add_E
rd_E
jumpOrBranch
pauseCore_E_o
cleanPipe_C_o
coldDownPipe_C_o
stallPipe_C_o
clear_MX
stall_MX
rd_M
regWrite_M
waitRequest_M_o
clear_W
stall_W
rd_W
regWrite_W
stageDEValid
stageDEMValid
AddressGeneration
changePCSrc_C_o_A_i
changePCSrc_E_o_A_i
nextPC_A_i
PCTarget_E_o_A_i
PCTarget_C_o_A_i
selectedPC_A_o
PC_A_o
changePCSrc
Register
enable
clear
Fetch
stall_F_i
instCountEn_F_o
stageSignalValid_F_o
instruction_F_o
disablePCAdder_F_i
PC_F_i
PC_F_o
32'd4
32'd2
32'd0
nextPC_F_o
instMemWaitRequest_F_i
instMemWaitRequest_F_o
InstructionLoadUnit
instructionMemoryData_F_i
instructionMemoryAddress_F_o
Decode
stall_D_i
instCountEn_D_i
stageSignalValid_D_i
instCountEn_D_o
stageSignalValid_D_o
mulEn_D_o
mulOpCode_D_o
pause_D_o
system_D_o
writeBackSrcSelect_D_o
branch_D_o
jump_D_o
regWrite_D_o
memRead_D_o
memWrite_D_o
ALUOpcode_D_o
ALUSrcAType_D_o
ALUSrcBType_D_o
funct3_D_o
control unit
immExtendSelect
instruction_D_i
Extend Unit
immExtend_D_o
PC_D_i
PC_D_o
nextPC_D_i
nextPC_D_o
funct12_D_o
rs1Data_D_i
rs1Data_D_o
rs2Data_D_i
rs2Data_D_o
rs1Addr_D_o
rs2Addr_D_o
rd_D_o
Multiplication1
mulEn_X1_i
mulOpCode_X1_i
multiplicationController
rs1Data_X1_i
rs2Data_X1_i
wallace_tree
mulEn_X1_o
wordSelect_X1_o
wallaceSum_X1_o
wallaceCarry_X1_o
Multiplication2
mulEn_X2_i
wordSelect_X2_i
wallaceSum_X2_i
wallaceCarry_X2_i
mulResult_X2_i
Execute
stall_E_i
changePCSrc_E_o_A_i
instCountEn_E_i
instCountEn_E_o
stageSignalValid_E_i
stageSignalValid_E_o
pauseUnit
pauseCore_E_o
pause_E_i
writeBackSrcSelect_E_i
writeBackSrcSelect_E_o
branch_E_i
jump_E_i
regWrite_E_i
regWrite_E_o
memRead_E_i
memRead_E_o
memWrite_E_i
memWrite_E_o
ALUOpcode_E_i
ALUSrcAType_E_i
ALUSrcBType_E_i
funct3_E_i
funct3_E_o
forwardA_E_i
forwardB_E_i
ALU
srcA
srcB
ALUResult_E_o
immExtend_E_i
PC_E_i
nextPC_E_i
nextPC_E_o
exeResult_M_o_E_i
writeBackResult_W_o_E_i
rs2Data_E_o
rs1Data_E_o
rs1Data_E_i
rs2Data_E_i
rs1Addr_E_i
rs2Addr_E_i
rd_E_i
rd_E_o
Memory
stall_M_i
instCountEn_M_i
instCountEn_M_o
stageSignalValid_M_i
stageSignalValid_M_o
writeBackSrcSelect_M_i
writeBackSrcSelect_M_o
regWrite_M_i
regWrite_M_o
memRead_M_i
memWrite_M_i
funct3_M_i
exeResult_M_i
nextPC_M_i
nextPC_M_o
exeResult_M_o
memDataOut_M_o
waitRequest_M_i
rs2Data_M_i
memDataOut_M_i
Load Store Unit
memWrite_M_o
memRead_M_o
waitRequest_M_o
memDataIn_M_o
memAccessType_M_o
memAddress_M_o
rd_M_i
rd_M_o
SelectWriteback
instCountEn_S_i
instCountEn_S_o
stageSignalValid_S_i
stageSignalValid_S_o
writeBackSrcSelect_S_i
regWrite_S_i
regWrite_S_o
mulResult_S_i
nextPC_S_i
exeResult_S_i
memDataOut_S_i
11
00
01
10
writeBackResult_S_o
rd_S_i
rd_S_o
Writeback
stall_W_i
instCountEn_W_i
instCountEn_W_o
stageSignalValid_W_i
stageSignalValid_W_o
regWrite_W_i
regWrite_W_o
writeBackResult_W_i
writeBackResult_W_o
rd_W_i
rd_W_o
RegisterFile
rs1Data_D_i
rs2Data_D_i
CSR Unit
funct3_C_i
cleanPipe_C_o
stallPipe_C_o
coldDownPipe_C_o
changePCSrc_C_o
changeExeSrc_C_o
PC_E_o_C_i
PC_A_o_C_i
rs1Data_C_i
rdAddr_C_i
csrAddr_C_i
funct12_C_i
rdData_C_o
PCTarget_C_o
InstructionMemoryReadRequest
InstructionMemoryAddress
InstructionMemoryData
InstructionMemoryWaitRequest
Instruction Memory
dataMemoryWriteData
dataMemoryAddress
dataMemoryReadRequest
dataMemoryWriteRequest
dataMemoryRequestMode
dataMemoryReadData
MemoryWaitRequest
Data Memory
read_data
wait_request
write_data


CSR Unit
CSRController
32'b0
CSRData
CSRWriteUnit
mvendorid
marchid
mimpid
mhartid
mconfigptr
mstatus
misa
mie
mtvec
mscratch
mepc
mcause
mtval
mip
mcpirate
mcpictrl
minstret
mcycle
mhpmcounter3
mhpmcounter4
mhpmcounter5
mhpmcounter6
mhpmcounter7
mhpmcounter8
mcountinhibit
CSRReadUnit
CSRTrapsPCGenerator
mtvec
mcause
mepc
return
targetPC
wfi
irq
CSRSleepingUnit
stallCore
priv
CPIRate
CPICTRL
CPIManagerUnit
CSRInterruptController
CSRTrapHandler
mret
ecall
raiseInterrupt
interruptCause
mstatus
PC_E_o
PC_A_o
changePCSrc
return
cleanPipe
mcauseEn
mcauseNewData
mtvalEn
mtvalNewData
mepcEn
mepcNewData
mstatusEn
mstatusNewData


CSRSleepingUnit
wfi
irq
stallCore


CSRSleepingUnit
pauseReq
pauseCore
carryOut | (~pauseReq)